MIL-E-63511(AR)
AMENDMENT 2
6 May 1986
SUPERSEDING
AMENDMENT 1
8 July 1985

MILITARY SPECIFICATION

ELECTRONICS-MCD ASSEMBLY
FOR
MINE, ANTI-TANK: BLU 91/B

This amendment forms a part of Military Specification MIL-E-63511(AR), dated 15 June 1982, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Preamble: Delete "Armament Research and Development Command" and substitute "Armament, Munitions and Chemical Command."

PAGE 2

SPECIFICATIONS, MILITARY:

MIL-S-19500/127: Delete in its entirety and add the following:

MIL-S-19500/444 - Semiconductor Device, Diode, Silicon, Switching Types 2N5711 and TX 2N5711"

PAGE 3

2.1.2, Other Government documents, drawings, and publications: US Army Armament Research and Development Command (ARRADCOM): Delete "Command (ARRADCOM)" and substitute "Center (ARDC)."

PAGE 4

3.4.1 Transportation vibration. Delete in its entirety and substitute the following:

"3.4.1 Transportation vibration. The Electronics-MCD Assembly shall be safe to handle and transport. Test mines shall show no evidence of damage that will affect the function following transportation vibration test No. 119, Procedure 2 of MIL-STD-331."

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* TABLE I:

Delete "Band Printed Wiring (Dwg. 9292230)" and substitute "Wiring, Printed Flexible (Dwg. 9292230)".

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4.4.2.1, Resistor, Fixed Composition, 5%, 1/8W:

TITLE: Delete "1/8W".

DRAWING NUMBER: Delete "MIL-R-39008/4" and substitute "MIL-R-39008/1/".

Part Number, Part Value: Delete in its entirety and substitute:

"Part Number 1/	Part Value
RCR05G 30 5 JR	3 M OHM
RCR05G 10 4 JR	100 K OHM
RCR05G 47 2 JR	4.7 K OHM
RCR05G 10 2 JR	1 K OHM
RCR07G 13 1 JR	130 K OHM"

Major 101: Delete "MIL-R-39008/4" and substitute "MIL-R-390088/1/".

Add the following note:

"1/ Use MIL-R-39008/1 or /4 as applicable, See Dwg. Note 13".

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4.4.2.4, Transistor, (P/N 9287122-1, JNTXV2N2907 and 9287122-2 JNTXV2N2222A):

Title: Delete "(P/N 9287122-1, JNTXV2N2907 and 9287122-2 JNTXU2N2222A)" and substitute "(JXTXV2907 and JNTXV2N2222A)"

Drawing Number: Delete "9287122" and substitute" MIL-S-19500/291N/255"

Major 101: Delete "Wire material 32 1/ 3.2 Visual" and substitute "Bond Wire Material not aluminum -- 3.2 Certification"

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Major 102: Delete in its entirety.

Major 105: Delete in its entirety.

Major 106: Delete in its entirety.

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4.4.2.5 Semi-conductor Device, Diode, Type JANTX IN 645-1:

Drawing Number: Delete "9287168-2" and substitute "MIL-S-19500/240"

Critical 1: Delete in its entirety.

Critical 2: Delete in its entirety.

Add the following Major Categories:

"104, Missing or incorrect color band on cathode end of diode - 100% 3.2 Electrical 1/"

"105, Reverse current - 100% 3.2 MIL-STD-750, Method 4016, dc"

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4.4.2.6, Diode, Zener, JANTX IN 4463 and JANTX IN 447l:

Critical 1: Delete in its entirety

Critical 2: Delete in its entirety

Critical 3: Delete in its entirety

Add the following Major Categories:

"104, Missing or incorrect color band on cathode end of diode - 100% 3.2 Electrical, 1/"

105, Reverse current - 100% 3.2 MIL-STD-750, Method 4016, dc"

106, Zener breakdown voltage V_Z - 100% 3.2 MIL-STD-750, Method 4022"

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4.4.2.7, Diode, Hot Carrier (Jan 1N 5711, MIL-S-19500/127): Change title to read "4.4.2.7, Diode, Hot Carrier (JAN 1N 5711, MIL-5-19500/444)"

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4.4.2.9, Capacitor, Fixed, Electrolytic Tantalum, Solid (82 uF):

Major 104: Delete "10-1-2" and substitute "10-2-3"

Major 106: Delete "8-0-1" and substitute "8-1-2"

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4.4.2.10, Capacitor, Fixed, Electrolytic Tantalum, Solid:

Major 104: Delete "10-1-2" and substitute "10-2-3"

Major 106: Delete "8-0-1" and substitute "8-1-2"

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4.4.2.16, Board, Printed Wiring:

Major 105: Delete "Large circular cutout, 2 places" and substitute "Diameter of circular cutout, 2 places".

Major 106: Delete "0.40%" and substitute "1/"

Add the following Note:

"1/ Printed Wiring Board sample plan per MIL-STD-105, AQL 0.40%, Inspection Level S4. From each board, randomly select and inspect 32 hole diameters. A board shall be classed defective if any hole is defective".

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4.4.2.1.8, Microcircuit, Digital, CMOS:

Major 101: Delete "1/" and substitute "5-1-2"

Note 1/: Delete in its entirety.

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4.4.2.24, Receiver Coil Assembly:

Major 104: Delete "Spacing of coil... 0.40% 3.2 Gage" and substitute "Bonding of coil to flexable circuit board incorrect 0.40% 3.2 Visual".

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* 4.4.2.26, Board, Printed Wiring:

TITLE: Delete "Board, Printed Wiring" and substitute "Wiring, Printed Flexible".

Major 103: Delete "Diameter" and substitute "Length and Width".

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4.4.2.29, Lid Electronics Assembly:

Minor 203: Delete in its entirety.

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4.4.2.3.5, Electronics-MCD Assy. (Prior to Assy of Test Mines):

Title: Delete "(Prior to Assy of Test Mines)"

Add the following Major Category:

"103 Improper soldering MCD flex cable - 0.40% 3.16 Visual"

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Add the following Major defect: "103 Impact failure 0-1 3.5 4.5.4"

PAGE 59

4.4.6 Launch Simulator

a. Delete "150 + 6.0" and substitute: "150 + 50"

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4.5.4.3, Disposition of Duds. Delete "ARRADCOM" and substitute "to the Government"

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In Table II Delete: "1. Shorting Bar Across Firing Capacitor 6 to 9 .2 to .7 OHMS" and substitute "1. Shorting Bar Across Firing Capacitor 6 to 9 .2 to .85 OHMS."

PAGE 68

5.1: Delete in its entirety and substitute the following:

"5.1 Packaging. The Electronics-MCD assembly shall be preserved and packaged in accordance with MIL-E-17555 with the following mandatory requirements:

Each individual Electronic-MCD Assembly shall be completely wrapped in metal foil (aluminum foil) to shield the units from electromagnetic and radio frequency energies.

The wrapped units shall be packaged in a corrugated fiberboard tray or box fitted with partitions and separators as required to keep the foil-wrapped units separated.

Cushioning as needed shall be used; the fiberboard trays or box shall be sealed in a barrier bag conforming to Tape I, or Class F, Style 1, Specification MIL-B-117. (Material used to fabricate the barrier bag must include metal foil).

The packaging materials shall not contain silicones, sulphur compounds or polysulphides."

6.3 Submission of inspection equipment designs for approval. Delete: "to: Commander, U.S. Army...New Jersey 07801" and substitute "to the Technical Agency specified in the contract."

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6.4 Submission of plans for approval. Delete "Commander: ARRADCOM, ... DRDAR-QAT-M" and substitute "the Technical Agency specified in the contract".

6.5 Submission of process control documents. Delete "to: Commander, ARRADCOM, ... DRDAR-QAT-M" and substitute "to the Technical Agency specified in the contract"

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6.6 Distribution of ammunition data cards, inspection and test results, process controls. Delete "Commander: ARRADCOM, ... DRDAR-QAT-M" and substitute "the Technical Agency specified in the contract"

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6.9 Definitions:

h. Transportation Vibration failure: Delete in its entirety and substitute the following.

"h. Transportation Vibration failure is defined as a test mine assembly that does not meet the requirements of 3.4.1, or in the event the borerider has functioned as a result of vibration the test shall be stopped and the Government Technical Agency specified in the Contract shall be notified for disposition of the test sample."

i. Impact failure: Delete in its entirety and substitute the following:

"i. Impact failure.

(1) Critical is defined as an Electronics-MCD Assembly that sustains damage due to impact resulting in critical failure modes for the MCD Assembly moves with respect to the Electronics Assembly (Dud failure analysis only).

(2) Major is defined as an Electronics-MCD Assembly that sustains damage due to impact resulting in a dud."

6.11 Drawings. Delete in its entirety and substitute:

"6.11 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as US Army Armament Research and Development Command, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC."

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